Machine Learning Quiz - Answers & Explanations

Q1. What is the primary reason to use machine learning instead of writing explicit rules for a problem?

Answer: B. It can discover patterns from data that are difficult to encode manually

Explanation: ML is used when rules are too complex/unknown to hand-code, so the model learns patterns directly from data.

Q2. A model achieves 98% training accuracy but only 82% testing accuracy. Most likely reason?

Answer: B. The model is overfitting the training data

Explanation: A large gap between training and testing accuracy means the model memorized training data patterns/noise instead of generalizing.

Q3. An email filter predicts 95% probability of spam. Best interpretation?

Answer: C. The model estimates a high likelihood that the email belongs to the spam class

Explanation: Predicted probabilities reflect the model's confidence/likelihood estimate for a class, not a guarantee or a measure of overall model accuracy.

Q4. A robot learns to navigate a maze via rewards/penalties. This is an example of:

Answer: C. Reinforcement learning

Explanation: Learning through rewards and penalties from interacting with an environment is the defining feature of reinforcement learning.

Q5. Small labeled data + large unlabeled data - which approach is most suitable?

Answer: A. Semi-supervised learning

Explanation: Semi-supervised learning combines a small labeled set with a large unlabeled set to improve learning beyond what labeled data alone allows.

Q6. Q1 = 10, Q3 = 20. Using the IQR method, what is the upper bound for outliers?

Answer: A. 35

Explanation: IQR = Q3 - Q1 = 20 - 10 = 10. Upper bound = Q3 + 1.5*IQR = 20 + 15 = 35.

Q7. What does R-squared measure in a regression model?

Answer: C. The variation in y explained by the model

Explanation: R-squared indicates the proportion of variance in the dependent variable that is explained by the independent variable(s).

Q8. Which step should normally occur before selecting a machine learning model?

Answer: D. Data preprocessing and exploration

Explanation: Understanding and cleaning the data (EDA, preprocessing) comes before model selection in a typical ML workflow.

Q9. Which method finds model parameters using a direct mathematical formula rather than iteration?

Answer: B. Closed Form Solution

Explanation: A closed-form solution (e.g., Normal Equation in linear regression) computes parameters directly via a formula, unlike iterative methods like gradient descent.

Q10. What is the goal of the cost function in linear regression?

Answer: D. Measure how well predictions match actual values

Explanation: The cost function quantifies the error between predicted and actual values, guiding parameter optimization to minimize this error.